

1, Explain behaviour of hub device

A hub is a basic networking device that works at the Physical layer. It receives data from one port and broadcasts it to all other ports, without checking the device's destination address.

Because of this, all connected devices receive the same data, which can cause collisions and makes the hub inefficient.

2, Explain behaviour of switch device

A switch is an intelligent networking device that works at the Data link layer. It reads the destination MAC Address in the frame and forwards it only to the correct port. This reduces unnecessary traffic, improves performance, and lowers the chance of collisions compared to a hub.

3, Explain behaviour of router device

A Router is a layer 3 device that connects different networks. It uses IP addresses and a routing table to choose the best path for forwarding packets between networks. Routers also separate broadcast domains, which makes them essential for communication between different LAN or Subnets.